

FC7xxx DIO User Manual

Rev A0

Revision History

Revision	Date	Changes
A0	2025/01/14	Initial release

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Chapter 1 DIO Introduction

1.1 Requirement Tracing

The design of this module follows the specifications of the DIO driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_SWS_DIODriver*.

1.2 Hardware Summary

The DIO Driver provides services for reading and writing to/from:

- DIO Channels (Pins)
- DIO Ports
- DIO Channel Groups

The behavior of those services is synchronous. This module works on pins and ports which are configured by the PORT driver for this purpose. For this reason, there is no configuration and initialization of this port structure in the DIO Driver.

The hardware configured by the Dio driver is GPIO.

The channel to microcontroller pin mapping can be done by using "FC7xxx Pinout.xlsx" from the Reference manual.

The value of an actual channel is identified by the formula:

$$\text{Channel} = \text{DioChannelId} + \text{DioPortId} * 32$$

Chapter 2 Software Design

2.1 Rejected Requirements

Rejected Requirement 1 **SWS_Dio_00083**

Description	If the microcontroller supports the direct read-back of a pin value, the Dio module's read functions shall provide the real pin level, when they are used on a channel that is configured as an output channel.
Rejection Reason	This works only if the output channel is configured as input, too

Rejected Requirement 2 **SWS_Dio_00084**

Description	If the microcontroller does not support the direct read-back of a pin value, the Dio module's read functions shall provide the value of the output register, when they are used on a channel that is configured as an output channel.
Rejection Reason	The read functions will only read the input registers, regardless of the channel configuration.

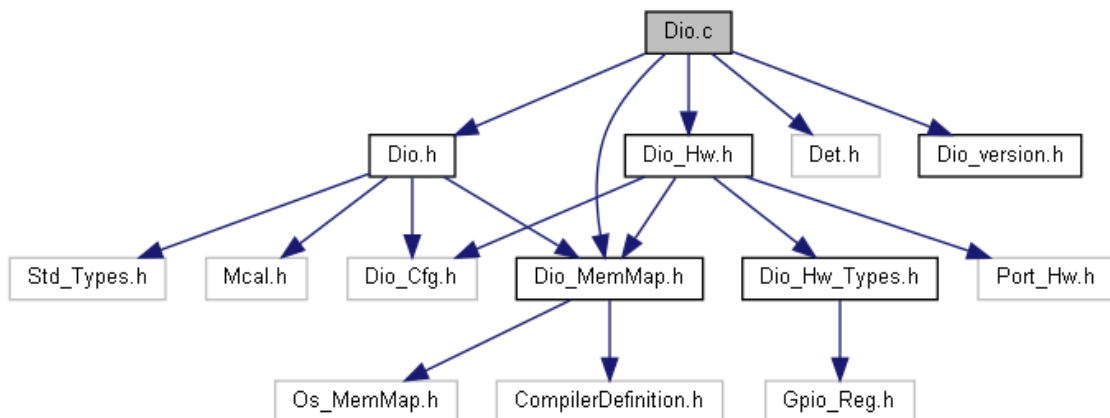
Rejected Requirement 3 **SWS_Dio_00104**

Description	When reading a port that is smaller than the Dio_PortType using the Dio_ReadPort function (see [SWS_Dio_00103]), the function shall set the bits corresponding to undefined port pins to 0. Furthermore, the requirements SWS_Dio_00005, SWS_Dio_00118 and SWS_Dio_00026 are applicable to the Dio_ReadPort function.
Rejection Reason	Requirement not applicable. Dio_ReadPort function always reads a port of exactly the size defined by Dio_PortType. It cannot read a port smaller than the size of the Dio_PortType.

Rejected Requirement 4 **SWS_Dio_00105**

Description	When writing a port that is smaller than the Dio_PortType using the Dio_WritePort function (see [SWS_Dio_00103]), the function shall ignore the MSB.
Rejection Reason	Requirement not applicable. Dio_WritePort function always writes a port of exactly the size defined by Dio_PortType. It cannot write a port smaller than the size of the Dio_PortType.

2.2 File Structure



2.3 Macros

2.3.1 Macros in Dio.h

- **#define DIO_E_PARAM_INVALID_CHANNEL_ID** ((uint8)0x0A)
Invalid channel name requested.
- **#define DIO_E_PARAM_CONFIG** ((uint8)0x10)
API service called with "NULL pointer" parameter.
- **#define DIO_E_PARAM_INVALID_PORT_ID** ((uint8)0x14)
Invalid port name requested.
- **#define DIO_E_PARAM_INVALID_GROUP_ID** ((uint8)0x1F)
Invalid ChannelGroup ID passed.
- **#define DIO_E_PARAM_POINTER** ((uint8)0x20)
API service called with a NULL pointer.
- **#define DIO_E_PARAM_LEVEL** ((uint8)0x21)
API service called with invalid channel level value.
- **#define DIO_READCHANNEL_ID** ((uint8)0x00)
API service ID for Dio_ReadChannel() function.
- **#define DIO_WRITECHANNEL_ID** ((uint8)0x01)
API service ID for Dio_WriteChannel() function.
- **#define DIO_FLIPCHANNEL_ID** ((uint8)0x11)
API service ID for Dio_FlipChannel() function.
- **#define DIO_READPORT_ID** ((uint8)0x02)
API service ID for Dio_ReadPort() function.
- **#define DIO_WRITEPORT_ID** ((uint8)0x03)
API service ID for Dio_WritePort() function.
- **#define DIO_READCHANNELGROUP_ID** ((uint8)0x04)
API service ID for Dio_ReadChannel() Group function.
- **#define DIO_WRITECHANNELGROUP_ID** ((uint8)0x05)
API service ID for Dio_WriteChannel() Group function.
- **#define DIO_GETVERSIONINFO_ID** ((uint8)0x12)
API service ID for DIO Get Version() Info function.

- **#define DIO_INSTANCE_ID** ((uint8)0x00)
Instance ID of the DIO driver.
- **#define DIO_CHECK_PORT_CHANNEL_VALID(ChannelId)** (u32ValidPortPins[ChannelId >> DIO_PORTID_SHIFT_U8] & (uint32) ((Dio_PortLevelType) 1 << ((ChannelId) & DIO_CHANNEL_MASK_U8)))
Check if the chip has the pin corresponding to ChannelId.

2.3.2 Macros in Dio_version.h

- **#define DIO_AR_RELEASE_MAJOR_VERSION** 4
- **#define DIO_AR_RELEASE_MINOR_VERSION** 6
- **#define DIO_AR_RELEASE_REVISION_VERSION** 0
- **#define DIO_MODULE_ID** 120
- **#define DIO_SW_MAJOR_VERSION** 1
- **#define DIO_SW_MINOR_VERSION** 1
- **#define DIO_SW_PATCH_VERSION** 0
- **#define DIO_VENDOR_ID** 174

2.3.3 Macros in Dio_Cfg.h

- **#define DIO_CFG_AR_RELEASE_MAJOR_VERSION** 4
- **#define DIO_CFG_AR_RELEASE_MINOR_VERSION** 6
- **#define DIO_CFG_AR_RELEASE_REVISION_VERSION** 0
- **#define DIO_CFG_SW_MAJOR_VERSION** 1
- **#define DIO_CFG_SW_MINOR_VERSION** 1
- **#define DIO_CFG_SW_PATCH_VERSION** 0
- **#define DIO_CFG_VENDOR_ID** 174
- **#define DIO_DEV_ERROR_DETECT** (STD_ON)
DIO Dev error detect switch.
- **#define DIO_VERSION_INFO_API** (STD_ON)
Function Dio_GetVersionInfo() enable switch.

- **#define DIO_FLIP_CHANNEL_API** (STD_ON)
Function Dio_FlipChannel() enable switch.
- **#define DIO_MASKED_WRITE_PORT_API** (STD_ON)
Function Dio_FlipChannel() enable switch.
- **#define DIO_NUM_PORTS_U16** (STD_ON)
Number of implemented ports.
- **#define DIO_NUM_CHANNELS_PER_PORT_U16** ((uint16)32)
Number channel in a port.
- **#define DIO_NUM_CHANNELS_U16** ((uint16)(DIO_NUM_PORTS_U16 * DIO_NUM_CHANNELS_PER_PORT_U16))
Number of channels available on the implemented ports.
- **#define DIO_MAX_VALID_OFFSET_U8** ((uint8)0x1F)
Mask representing the maximum valid offset for a channel group.
- **#define DIO_PRECOMPILE_SUPPORT**
DIO driver Pre-Compile configuration switch.
- **#define DIO_ENABLE_USER_MODE_SUPPORT** (STD_ON)
Support for User mode.
- **#define MAX_CONFIGURED_DIO_CHANNEL_GROUPS** ((uint8)1)
The number of DIO channel groups configured in configuration tools.
- **#define s_DioConfigChannels** (Dio_Config)
Symbolic name for the configuration s_DioConfigChannels.
- **#define DioConf_DioPort_DioPort_PTA** ((uint8)0x00U)
Symbolic name for the port DioPort_PTA.
- **#define DioConf_DioChannel_Dio_PTA11** ((uint16)0x000bU)
Symbolic name for the channel Dio_PTA11.

2.4 Enums

N/A

2.5 Typedefs

2.5.1 Typedefs in Dio_Hw_Types.h

- **typedef uint8 Dio_PortType**
Numeric ID of a DIO port.

- typedef uint16 **Dio_ChannelType**
Numeric ID of a DIO channel.
- typedef uint32 **Dio_PortLevelType**
Type of a DIO port levels representation.
- typedef uint8 **Dio_LevelType**
Type of a DIO channel levels representation.

2.6 Structures

2.6.1 Dio_ChannelGroupType

Structure	Dio_ChannelGroupType
Description	The structure indicates the information of a single DIO channel group.
Diagram	N/A
Data Fields	• Dio_PortType port This shall be the port on which the Channel group is defined. • uint8 offset This element shall be the position of the Channel Group on the port, counted from the LSB. • Dio_PortLevelType mask This element mask which defines the positions of the channel group.

2.6.2 Dio_ConfigType

Structure	Dio_ConfigType
Description	This structure contains the information of all channel groups.
Diagram	<pre> graph TD DC[Dio_ConfigType] -- solid --> DCS[Dio_ChannelGroupType_struct] DC -. dashed pChannelGroupList .-> DCG[Dio_ChannelGroupType] DCG -- solid --> DCS </pre> [legend]
Data Fields	• const Dio_ChannelGroupType *pChannelGroupList Pointer containing the information of all channel groups.

2.6.3 GPIO_Type

Structure	GPIO_Type
Description	Register Layout Typedef
Diagram	N/A
Data Fields	• volatile uint32 PDOR Port Data Output Register, offset: 0x0 • volatile uint32 PSOR Port Set Output Register, offset: 0x4

	- volatile uint32 PCOR Port Clear Output Register, offset: 0x8 - volatile uint32 PTOR Port Toggle Output Register, offset: 0xC - volatile uint32 PDIR Port Data Input Register, offset: 0x10 - volatile uint32 PDDR Port Data Direction Register, offset: 0x14 - volatile uint32 PIDR Port Input Disable Register, offset: 0x18
--	---

2.7 API Functions

2.7.1 Functions in Dio.h

2.7.1.1 Dio_LevelType Dio_ReadChannel(const Dio_ChannelType ChannelId)

Function	Dio_LevelType Dio_ReadChannel(const Dio_ChannelType ChannelId)	
Description	Returns the value of the specified DIO channel.	
Diagram	 <pre> graph LR A[Dio_ReadChannel] --> B[Dio_HL_ReadChannel] </pre>	
Parameters	Parameter	Description
	ChannelId	ID of DIO channel.
Returns	Dio_LevelType	

2.7.1.2 void Dio_WriteChannel(const Dio_ChannelType ChannelId, const Dio_LevelType Level)

Function	void Dio_WriteChannel(const Dio_ChannelType ChannelId, const Dio_LevelType Level)	
Description	Service to set a level of a channel.	
Diagram	 <pre> graph LR A[Dio_WriteChannel] --> B[Dio_HL_WriteChannel] </pre>	
Parameters	Parameter	Description
	ChannelId	ID of DIO channel.
	Level	Value to be written.
Returns	N/A	

2.7.1.3 Dio_LevelType Dio_FlipChannel(const Dio_ChannelType ChannelId)

Function	Dio_LevelType Dio_FlipChannel(const Dio_ChannelType ChannelId)	
Description	Service to flip (change from 1 to 0 or from 0 to 1) the level of a channel and return the level of the channel after flip.	
Diagram	 <pre> graph LR A[Dio_FlipChannel] --> B[Dio_HL_FlipChannel] </pre>	
Parameters	Parameter	Description
	ChannelId	ID of DIO channel.
Returns	Dio_LevelType	

2.7.1.4 Dio_PortLevelType Dio_ReadPort(const Dio_PortType PortId)

Function	Dio_PortLevelType Dio_ReadPort(const Dio_PortType PortId)	
Description	Returns the level of all channels of that port.	
Diagram	 <pre> graph LR A[Dio_ReadPort] --> B[Dio_HL_ReadPort] </pre>	

Parameters	Parameter	Description
	PortId	ID of DIO Port.
Returns	Dio_PortLevelType	

2.7.1.5 void Dio_WritePort(const Dio_PortType PortId, const Dio_PortLevelType Level)

Function	void Dio_WritePort(const Dio_PortType PortId, const Dio_PortLevelType Level)	
Description	Service to set a value of the port.	
Diagram	 <pre> graph LR A[Dio_WritePort] --> B[Dio_HL_WritePort] </pre>	
Parameters	Parameter	Description
	PortId	ID of DIO Port.
	Level	Value to be written.
Returns	N/A	

2.7.1.6 Dio_PortLevelType Dio_ReadChannelGroup(const Dio_ChannelGroupType *ChannelGroupIdPtr)

Function	Dio_PortLevelType Dio_ReadChannelGroup(const Dio_ChannelGroupType *ChannelGroupIdPtr)	
Description	This service reads a subset of the adjoining bits of a port.	
Diagram	 <pre> graph LR A[Dio_ReadChannelGroup] --> B[Dio_HL_ReadChannelGroup] </pre>	
Parameters	Parameter	Description
	ChannelGroupIdPtr	Pointer to the channel group.
Returns	Dio_PortLevelType	

2.7.1.7 void Dio_WriteChannelGroup(const Dio_ChannelGroupType *ChannelGroupIdPtr, const Dio_PortLevelType Level)

Function	void Dio_WriteChannelGroup(const Dio_ChannelGroupType *ChannelGroupIdPtr, const Dio_PortLevelType Level)	
Description	Service to set a subset of the adjoining bits of a port to a specified level.	
Diagram	 <pre> graph LR A[Dio_WriteChannelGroup] --> B[Dio_HL_WriteChannelGroup] </pre>	
Parameters	Parameter	Description
	ChannelGroupIdPtr	Pointer to the channel group.
	Level	Value to be written.
Returns	N/A	

2.7.1.8 void Dio_GetVersionInfo(Std_VersionInfoType *VersionInfo)

Function	void Dio_GetVersionInfo(Std_VersionInfoType *VersionInfo)	
Description	Service to get the version information of this module.	
Parameters	Parameter	Description
	versioninfo	Pointer to where to store the version information of this module.
Returns	N/A	

2.7.1.9 void Dio_MaskedWritePort(Dio_PortType PortId, Dio_PortLevelType Level, Dio_PortLevelType Mask)

Function	void Dio_MaskedWritePort(Dio_PortType PortId, Dio_PortLevelType Level, Dio_PortLevelType Mask)	
Description	Service to get the version information of this module.	
Diagram	 <pre> graph LR A[Dio_MaskedWritePort] --> B[Dio_HL_WritePort] </pre>	

Parameters	Parameter	Description
	PortId	ID of DIO Port.
	Level	Value to be written
	Mask	Channels to be masked in the port
Returns	N/A	

2.8 Hardware Functions

2.8.1 Functions in Dio_Hw.h

2.8.1.1 Dio_LevelType Dio_HL_ReadChannel(const Dio_ChannelType u16ChannelId)

Function	Dio_LevelType Dio_HL_ReadChannel(const Dio_ChannelType u16ChannelId)	
Description	Reads the channel level.	
Parameters	Parameter	Description
	u16ChannelId	ID of the DIO channel.
Returns	Std_ReturnType Function return type.	
Referenced By	Dio_ReadChannel()	

2.8.1.2 void Dio_HL_WriteChannel(const Dio_ChannelType u16ChannelId, const Dio_LevelType u8Level)

Function	void Dio_HL_WriteChannel(const Dio_ChannelType u16ChannelId, const Dio_LevelType u8Level)	
Description	Writes the level to the channel.	
Parameters	Parameter	Description
	u16ChannelId	ID of the DIO channel.
Returns	N/A	
Referenced By	Dio_WriteChannel	

2.8.1.3 Dio_LevelType Dio_HL_FlipChannel(const Dio_ChannelType u16ChannelId)

Function	Dio_LevelType Dio_HL_FlipChannel(const Dio_ChannelType u16ChannelId)	
Description	Flips the level of the channel.	
Parameters	Parameter	Description
	u16ChannelId	ID of the DIO channel.
Returns	Dio_LevelType	
Referenced By	Dio_FLipChannel()	

2.8.1.4 Dio_PortLevelType Dio_HL_ReadPort(const Dio_PortType u8PortId)

Function	Dio_PortLevelType Dio_HL_ReadPort(const Dio_PortType u8PortId)	
Description	Reads the level of Port.	
Parameters	Parameter	Description
	u8PortId	ID of the DIO Port.
Returns	Dio_PortLevelType	
Referenced By	Dio_ReadPort()	

2.8.1.5 void Dio_HL_WritePort(const Dio_PortType u8PortId, const Dio_PortLevelType u32Level)

Function	void Dio_HL_WritePort(const Dio_PortType u8PortId, const Dio_PortLevelType u32Level)	
Description	Writes the port level.	
Parameters	Parameter	Description

	u8PortId	ID of the DIO Port.
	u32Level	Value to be written.
Returns	N/A	
Referenced By	Dio_WritePort()	

2.8.1.6 Dio_PortLevelType Dio_HL_ReadChannelGroup(const Dio_ChannelGroupType *pChannelGroupIdPtr)

Function	Dio_PortLevelType Dio_HL_ReadChannelGroup(const Dio_ChannelGroupType *pChannelGroupIdPtr)	
Description	Reads the level of the channels group.	
Parameters	Parameter	Description
	pChannelGroupIdPtr	Channel group information pointer.
Returns	Dio_PortLevelType	
Referenced By	Dio_ReadChannelGroup()	

2.8.1.7 void Dio_HL_WriteChannelGroup(const Dio_ChannelGroupType *pChannelGroupIdPtr, const Dio_PortLevelType u32Level)

Function	void Dio_HL_WriteChannelGroup(const Dio_ChannelGroupType *pChannelGroupIdPtr, const Dio_PortLevelType u32Level)	
Description	Writes the level of the channels group.	
Parameters	Parameter	Description
	pChannelGroupIdPtr	Channel group information pointer.
	u32Level	Value to be written.
Returns	N/A	
Referenced By	Dio_WriteChannelGroup()	

2.8.1.8 Dio_HL_ValidatePinAuth(const Dio_ChannelType u16ChannelId, const Dio_CoreConfigInfoType pCoreConfigInfo[])

Function	Dio_HL_ValidatePinAuth(const Dio_ChannelType u16ChannelId, const Dio_CoreConfigInfoType pCoreConfigInfo[])	
Description	Reads the level of the channels group.	
Parameters	Parameter	Description
	u16ChannelId	Id of the Dio channel
	pCoreConfigInfo	Config info of the channel partition
Returns	Std_ReturnType	
Referenced By	Dio_ReadChannelGroup()	

2.8.1.9 Dio_HL_ValidatePortAuth (const Dio_PortType u8PortId, const Dio_PortLevelType u32Mask)

Function	Dio_HL_ValidatePortAuth (const Dio_PortType u8PortId, const Dio_PortLevelType u32Mask)	
Description	Reads the level of the channels group.	
Parameters	Parameter	Description
	u8PortId	Id of the Port
	u32Mask	Mask of the pins to be written
Returns	Std_ReturnType	
Referenced By	Dio_ReadChannelGroup()	

2.9 Peripheral Functions

2.9.1 Functions in Dio_RegOps.h

2.9.1.1 LOCAL_INLINE Dio_PortLevelType Dio_HWA_ReadPortDataInput(GPIO_Type *pGpio)

Function	LOCAL_INLINE Dio_PortLevelType Dio_HWA_ReadPortDataInput(GPIO_Type *pGpio)	
Description	Reads data on Port.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
Returns	Dio_PortLevelType	
Referenced By	Dio_HL_ReadChannel(), Dio_HL_ReadPort(), Dio_HL_ReadChannelGroup()	

2.9.1.2 LOCAL_INLINE void Dio_HWA_ClearPinOutput(GPIO_Type *pGpio, uint8 u8Pin)

Function	LOCAL_INLINE void Dio_HWA_ClearPinOutput(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Sets the pin output to 0.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Dio_HL_WriteChannel()	

2.9.1.3 LOCAL_INLINE void Dio_HWA_SetPinDataOutput(GPIO_Type *pGpio, uint8 u8Pin)

Function	LOCAL_INLINE void Dio_HWA_SetPinDataOutput(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Sets the pin data output. The pin output can be set to 1 or 0.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Dio_HL_WriteChannel()	

2.9.1.4 LOCAL_INLINE void Dio_HWA_TogglePin(GPIO_Type *pGpio, uint8 u8Pin)

Function	LOCAL_INLINE void Dio_HWA_TogglePin(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Toggles pin.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	

2.9.1.5 LOCAL_INLINE Dio_PortLevelType Dio_HWA_ReadPortDataOutput(GPIO_Type *pGpio)

Function	LOCAL_INLINE Dio_PortLevelType Dio_HWA_ReadPortDataOutput(GPIO_Type *pGpio)	
Description	Reads the port data output.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
Returns	Dio_PortLevelType	
Referenced By	Dio_HL_FlipChannel()	

2.9.1.6 LOCAL_INLINE void Dio_HWA_SetPortDataOutput(GPIO_Type *pGpio, uint32 u32Pins)

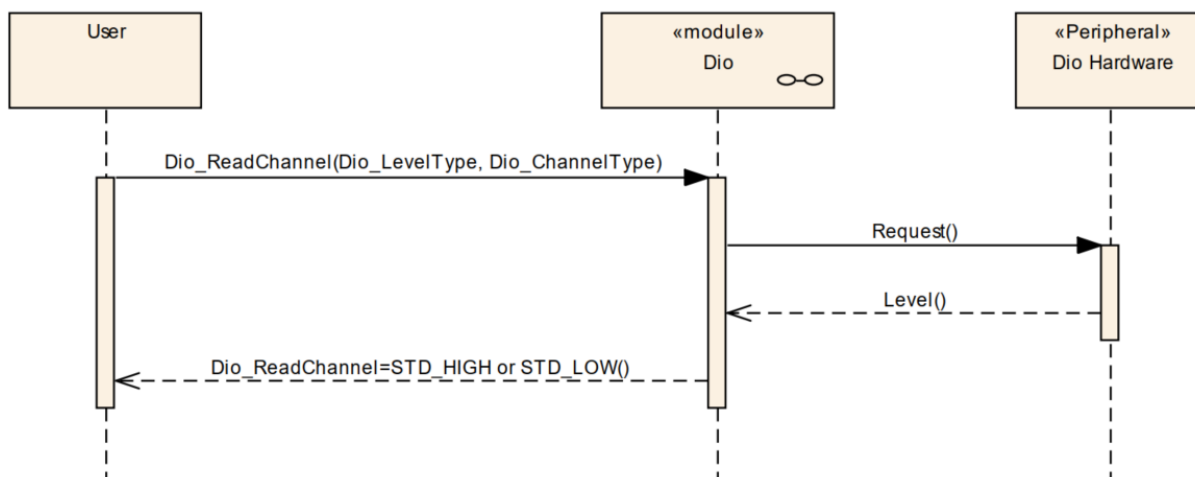
Function	LOCAL_INLINE void Dio_HWA_SetPortDataOutput(GPIO_Type *pGpio, uint32 u32Pins)	
Description	Sets the port data output. The port output can be set to 1 or 0.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
	u32Pins	Pin numbers.
Returns	N/A	
Referenced By	Dio_HL_WritePort()	

2.9.1.7 LOCAL_INLINE void Dio_HWA_SetPortOutput(GPIO_Type *pGpio, uint32 u32Pins)

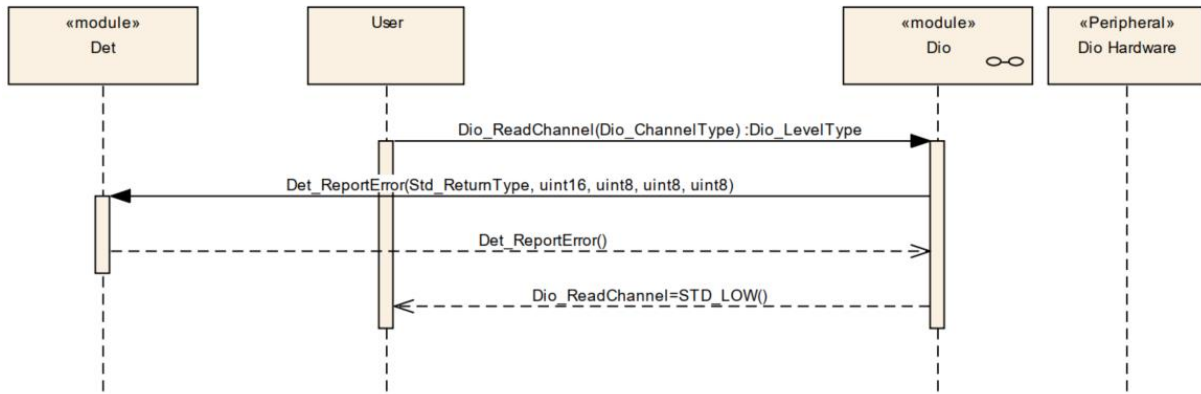
Function	LOCAL_INLINE void Dio_HWA_SetPortOutput(GPIO_Type *pGpio, uint32 u32Pins)	
Description	Sets the port output to 1.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
	u32Pins	Pin numbers.
Returns	N/A	
Referenced By	Dio_HL_WriteChannelGroup()	

2.9.1.8 LOCAL_INLINE void Dio_HWA_ClearPortOutput(GPIO_Type *pGpio, uint32 u32Pins)

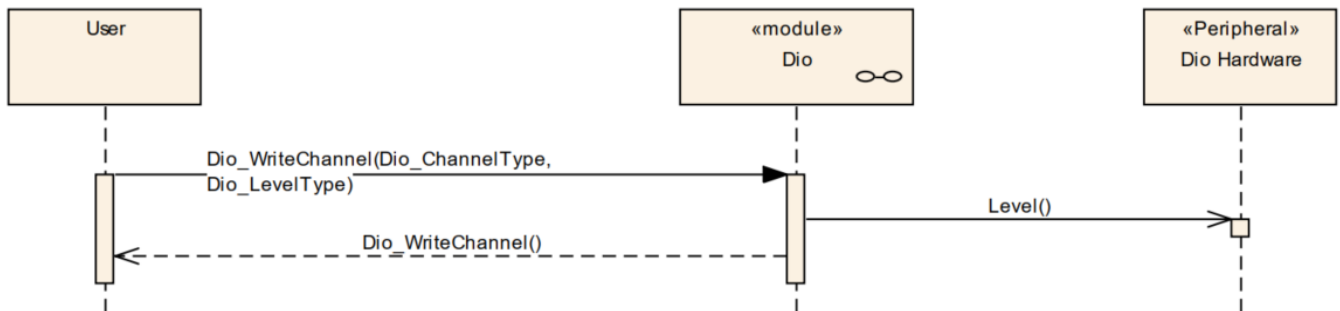
Function	LOCAL_INLINE void Dio_HWA_ClearPortOutput(GPIO_Type *pGpio, uint32 u32Pins)	
Description	Sets the port output to 0.	
Parameters	Parameter	Description
	pGpio	GPIO instance.
	u32Pins	Pin numbers.
Returns	N/A	
Referenced By	Dio_HL_WriteChannelGroup()	

2.10 API Sequence Diagram**2.10.1 Read Value from Digital I/O – 1**

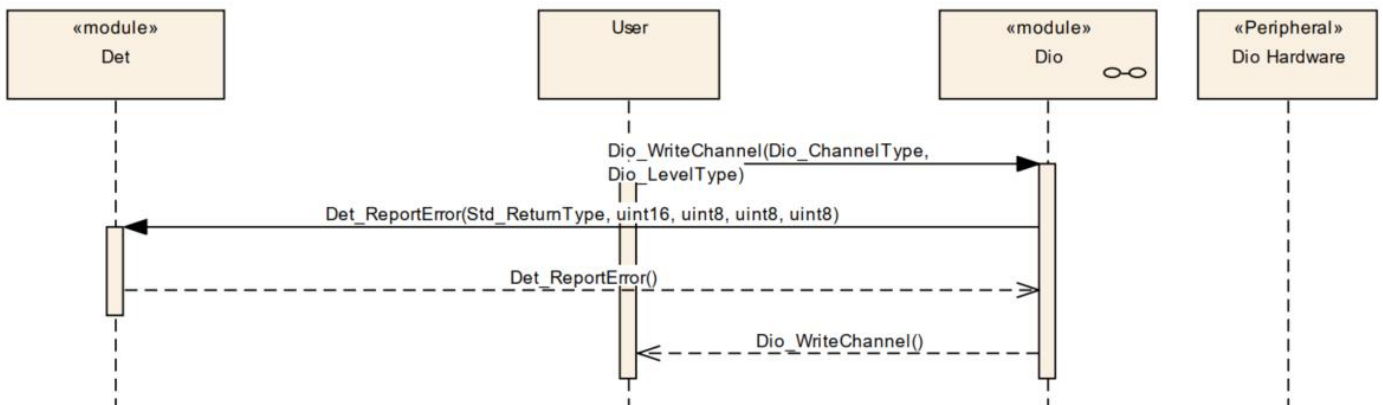
2.10.2 Read Value from Digital I/O – 2



2.10.3 Write Value to Digital I/O - 1



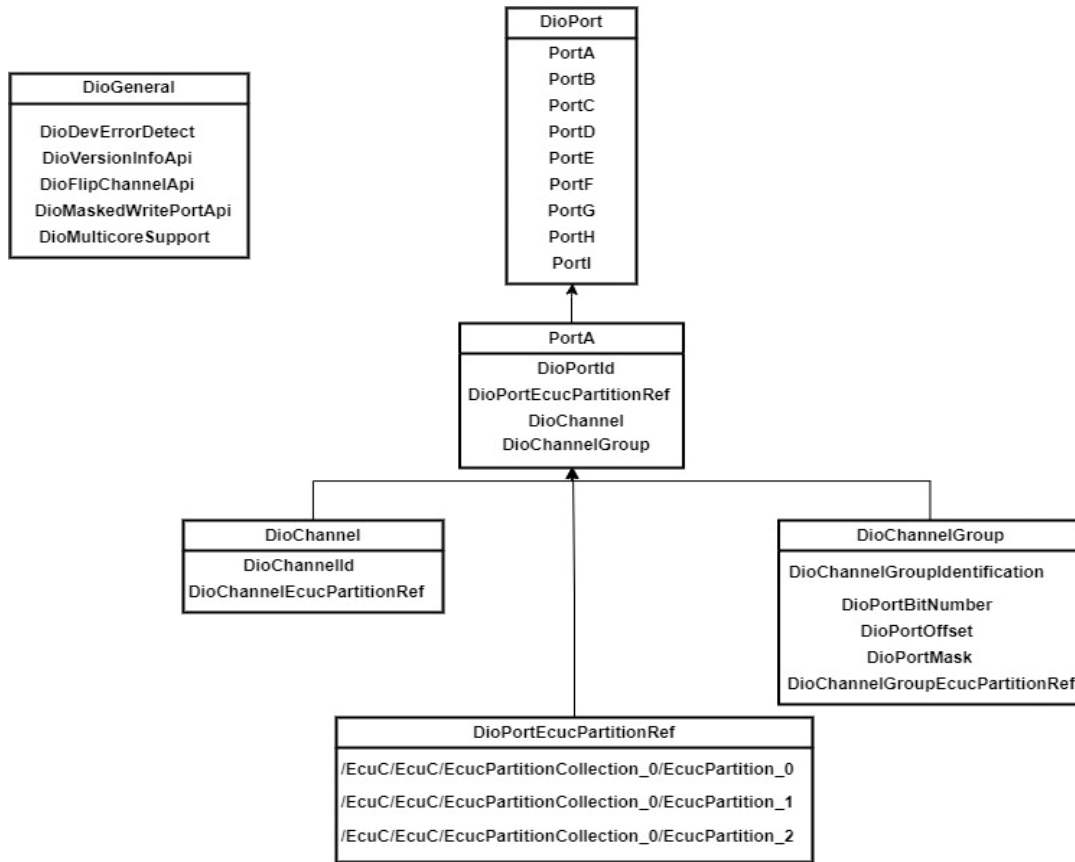
2.10.4 Write Value to Digital I/O - 2



Chapter 3 Tresos Configuration Items

3.1 Container Inclusion Relation

The container inclusion relation is shown as below:



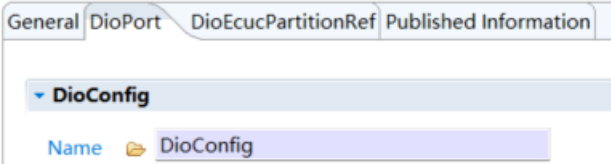
3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

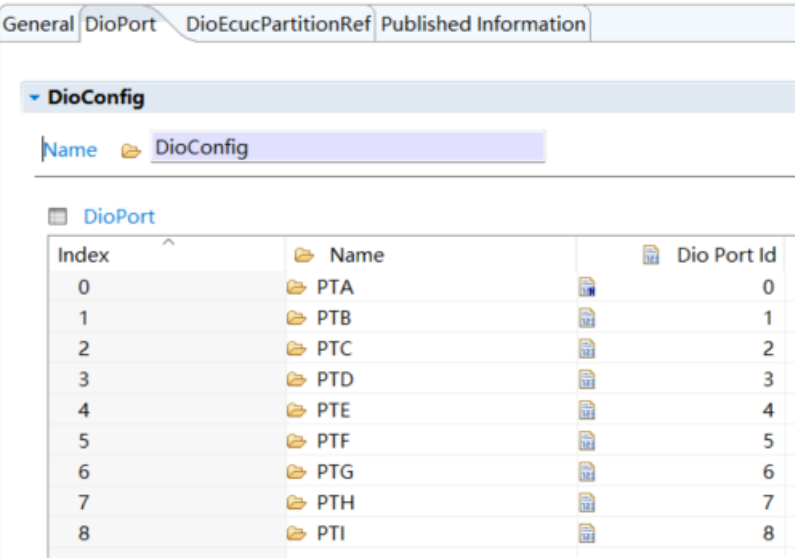
Variant	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable: enumeration
	Label	Config Variant
	Default	VariantPrecompile
	Range	VariantLinkTime VariantPreCompile
Returns	N/A	

3.2.2 DioConfig

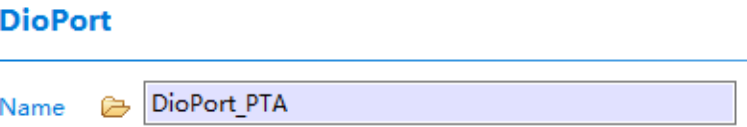
Container	DioConfig
Description	This container contains the configuration parameters and sub containers of the AUTOSAR DIO module. This container is a MultipleConfigurationContainer, i.e., this container and its sub-containers exist once

	per configuration set.	
Screenshot		
Properties	Property	Value
	Type	Container
	TAB	DioPort

3.2.2.1 DioPort

List	DioPort	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Map
	Min	1

3.2.2.1.1 DioPort

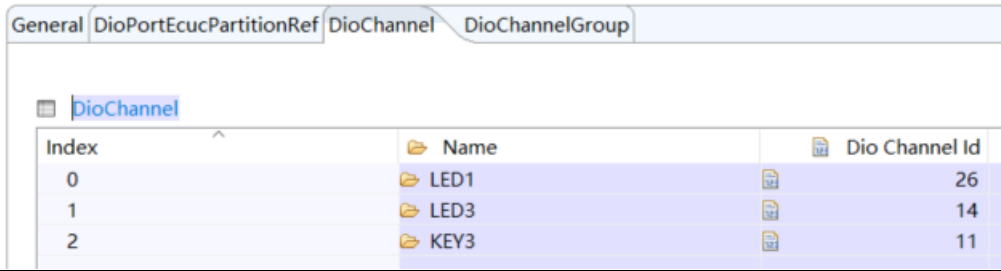
Container	DioPort	
Description	Configuration of individual DIO ports, consisting of channels and possible channel groups.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1.2 DioPortId

Variable	DioPortId	
Description	Numeric identifier of the DIO port. Symbolic names will be generated for each port pin id for the pins being used for configuration. Use the following values to configure different ports. PortA = 0, PortB = 1, PortC = 2, PortD = 3, PortE = 4, PortF = 5, PortG = 6, PortH = 7, PortI = 8.	
Screenshot		

Properties	Property	Value
	Type	Variable:Integer
	Label	Dio Port Id
	Default	0
	Invalid	0~4
	Range	1. Cannot be repeated 2. The selected Port must have a valid Pin (Dio.ValidPortPins cannot be 0 in the resource file).

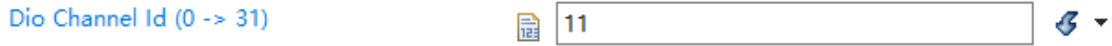
3.2.2.1.3 DioChannel

List	DioChannel	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	MAP

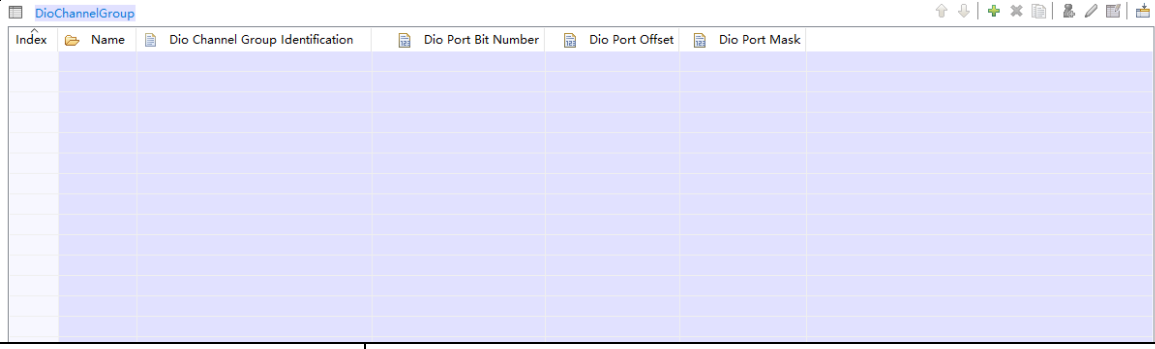
3.2.2.1.4 DioChannel

Container	DioChannel	
Description	Configuration of an individual DIO channel. Symbolic names will be generated for each channel. A general-purpose digital IO pin represents a DIO channel that will be having a value of either STD_HIGH or STD_LOW.	
Screenshot		
Properties	Property	Value
	Type	Container

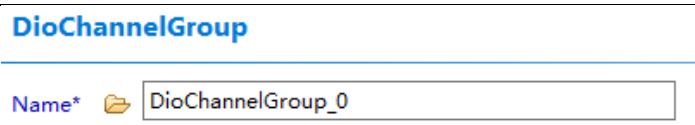
3.2.2.1.5 DioChannelId

Variable	DioChannelId	
Description	Channel Id of the DIO channel. This value will be assigned to the symbolic names.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	True
	Default	0
	Invalid	1. Multiple ChannelIDs are duplicated 2. This channelId is not a ValidPins
	Range	>=0, <=31.

3.2.2.1.6 DioChannelGroup

List	DioChannelGroup	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	MAP
	Max	255

3.2.2.1.7 DioChannelGroup

Container	DioChannelGroup	
Description	A channel group represents several adjoining DIO channels represented by a logical group. This container definition does not explicitly define a symbolic name parameter, but symbolic names will be generated for each channel group. Each group provides a structure with parameters port, offset and mask.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1.8 DioChannelGroupIdentification

Variable	DioChannelGroupIdentification	
Description	A DIO channel group is identified in DIO APIs by a pointer to a data structure of type Dio_ChannelGroupType. This data structure contains channel group information. This parameter contains the code fragment that has to be inserted in the API call of the calling module to get the address of the variable in memory which holds the channel group information, a string value should be given for this parameter. Symbolic names will be generated for each DioChannelGroup, which will be assigned with address of this string in order to point to the structure parameters. Example: OutputGroup	
Screenshot		
Properties	Property	Value
	Type	Variable: String
	Label	Dio Channel Group Identification
	Origin	AUTOSAR_ECUC
	Default	DioChannelGroup

3.2.2.1.9 DioPortBitNumber

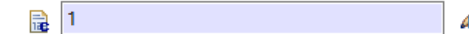
Variable	DioPortBitNumber
Description	This is the number of continuous channels that create a channel group.

Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	1

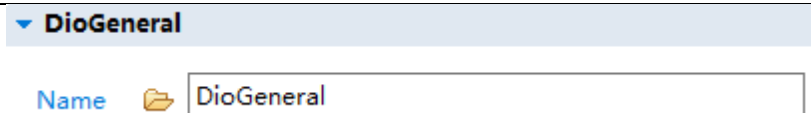
3.2.2.1.10 DioPortOffset

Variable	DioPortOffset	
Description	The position of the Channel Group on the port, counted from the least significant bit. This value can be derived from DioPortMask.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Label	Dio Port Offset
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0

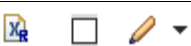
3.2.2.1.11 DioPortMask

Variable	DioPortMask	
Description	The mask defines the positions of the channel group.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	Calculated from "Dio Port Bit Number" and "Dio Port Offset" automatically.

3.2.3 DioGeneral

Container	DioGeneral	
Description	General DIO module configuration parameters.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.1 DioDevErrorDetect

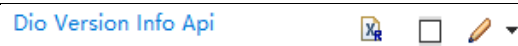
Variable	DioDevErrorDetect	
Description	Switches the Development Error Detection and Notification ON or OFF.	
Screenshot		
Properties	Property	Value
	Type	Variable:Boolean

	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.2 DioVersionInfoApi

Variable	DioVersionInfoApi	
Description	Adds/removes the service Dio_GetVersionInfo() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable:Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

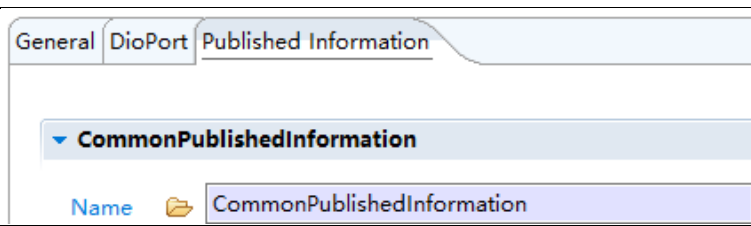
3.2.3.3 DioFlipChannelApi

Variable	DioFlipChannelApi	
Description	Adds/removes the service Dio_FlipChannel() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable:Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.4 DioMulticoreSupport

Variable	DioMulticoreSupport	
Description	This parameter is added in Dio configuration in order to enable multicore mode.	
Screenshot		
Properties	Property	Value
	Type	Variable:Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.4 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	N/A	

3.2.4.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot	ArReleaseMajorVersion  4	
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	4

3.2.4.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot	AUTOSAR Minor Version  6	
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	6

3.2.4.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot	AUTOSAR Release Revision Version  0	
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.4.4 SwMajorVersion

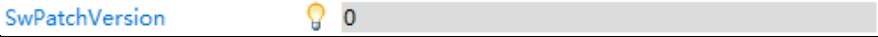
Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot	SwMajorVersion  0	
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.4.5 SwMinorVersion

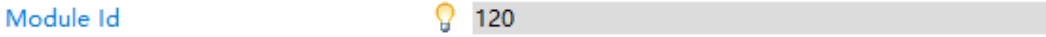
Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	

	specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	3

3.2.4.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.4.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	120

3.2.4.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	174

Chapter 4 Configuration Guides

4.1 DIO Usage Common Steps

- Dio ports:

In order to use Dio port APIs (Dio_ReadPort(), Dio_WritePort()) on the Port you need. Here are some suggested steps:

- 1) Ensure which port you want to use (e. g. PORT B).
- 2) Go to tab "DioPort", add a new Port, the Dio Port Id is the numeric identifier of the port you want to use(e. g. 1 corresponds to PORT B, 4 corresponds to PORTE).
- 3) Generate code.
- 4) Open Dio_Cfg.h file and look inside the "DEFINES AND MACROS" section for the symbolic name of the Dio Port. Use these symbolic name macros as parameters when calling Dio port APIs.

- Dio channels:

In order to use Dio channel APIs (Dio_ReadChannel(), Dio_WriteChannel (),Dio_FlipChannel ()) on the microcontroller hardware pin you need.. Here are some suggested steps:

- 1) Ensure which Pin you want to use.
- 2) Go to tab "DioPort", find the corresponding Port and double-click its index.
- 3) Go to tab "DioChannel", add a new Dio channel to container "DioChannel". Set "Dio Channel Id" to the corresponding number of the Pin you want to use.
- 4) Generate code.
- 5) Open Dio_Cfg.h file and look inside the "DEFINES AND MACROS" section for the symbolic name of the Dio Channel. Use these symbolic name macros as parameters when calling Dio channel APIs.

- Dio channel groups:

In order to use Dio channel group APIs (Dio_ReadChannelGroup(), Dio_WriteChannelGroup()) on several adjoining Dio channels that belong to one Dio port. Here are some suggested steps:

- 1) Ensure the Dio channel group you want to use (e. g. PTE4/PTE5/PTE6).
- 2) Go to tab "DioPort", find the corresponding Port and double click its index.
- 3) Go to tab "DioChannelGroup", add a new Dio channel group in container "DioChannelGroup".
- 4) Calculate the "Dio Port Bit Number", "Dio Port Offset" according to the Dio channel group you want to use. (e. g. For channel group "PTE4/PTE5/PTE6", "Dio Port Bit Number" should be 3, "Dio Port Offset" should be 4)
- 5) Click the "calculate value" button, the value of "Dio Port Mask" will be calculated automatically.
- 6) Generate code.
- 7) Open Dio_Cfg.h file and look inside the "DEFINES AND MACROS" section for the symbolic name of the Dio Channel Groups. Use these symbolic name macros as parameters when calling Dio channel Group APIs.